

Gesture Recognition

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Introduction

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MediaPipe is an open-source, data-stream processing machine learning application development framework developed and open-sourced by Google. It is a graph-based data processing pipeline used to build applications that consume multiple forms of data sources, such as video, audio, sensor data, and any time-series data.

MediaPipe is cross-platform and can run on embedded platforms (such as Raspberry Pi), mobile devices (iOS and Android), workstations, and servers, and it supports mobile GPU acceleration. MediaPipe provides cross-platform, customizable ML solutions for real-time and streaming media.

The core framework of MediaPipe is implemented in C++, with support for languages such as Java and Objective C. The main concepts of MediaPipe include Packet, Stream, Calculator, Graph, and Subgraph.

Features of MediaPipe:

- End-to-end acceleration: Built-in fast ML inference and processing can accelerate even on ordinary hardware.
- Build once, deploy anywhere: Unified solutions work on Android, iOS, desktop/cloud, web, and IoT.
- Ready-to-use solutions: Cutting-edge ML solutions that showcase the full functionality of the framework.
- Free and open source: Framework and solutions under Apache 2.0, fully extensible and customizable.

Gesture Recognition

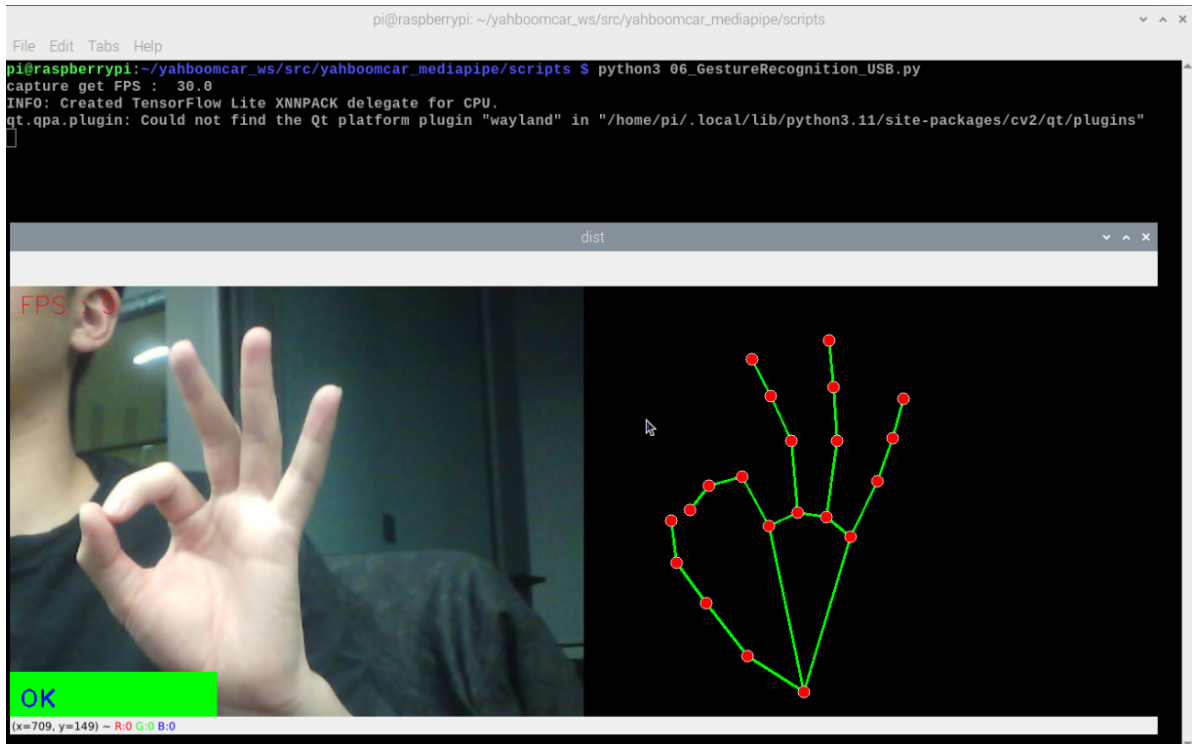
Source code location: /home/pi/yahboomcar_ws/src/yahboomcar_mediapipe/scripts

The gesture recognition is designed based on the right hand. When specific conditions are met, it can be recognized accurately. The recognizable gestures are: [Zero, One, Two, Three, Four, Five, Six, Seven, Eight, Ok, Rock, Thumb_up (like), Thumb_down (thumbs down), Heart_single (one-hand heart)], 14 categories in total.

If you want to exit the program, you can press q in the preview window or press Ctrl+C in the terminal to terminate the program!

1. USB Camera

```
cd /home/pi/yahboomcar_ws/src/yahboomcar_mediapipe/scripts
python3 06_GestureRecognition_USB.py
```



2. CSI Camera

```
cd /home/pi/yahboomcar_ws/src/yahboomcar_mediapipe/scripts
python3 06_GestureRecognition_CSI.py
```

